

DATA602 Final Group Project Checkpoint 1

Project Name: Austin Housing Price Prediction

Group members: Jed Daleiden, Elliot Fomin, Hajime Inoue, Michael McDonald

Topic

How accurately can we predict house prices in the Austin, TX area based on property characteristics and location?

Github Repository

<https://github.com/Mhycka1/DATA602-Group-Project>

Dataset

Austin, TX House Listings by Eric Pierce

<https://www.kaggle.com/datasets/ericpierce/austinhousingprices>

This dataset includes 15,171 different houses within the Austin, TX area listing 47 different features for those houses. The different cities include West Lake Hills, Dripping Springs, Driftwood, Austin, Manor, Manchaca, Pflugerville, and Del Valle which are further broken apart by zip code and latitude/longitude. The dataset includes both categorical and numerical features to aid in trying to find a relationship between the features and the target variable, 'latestPrice'.

Why are we choosing this dataset?

Analyzing this dataset can provide insights into which characteristics are most valuable for homes in the Austin, Texas area which is a budding market for real estate in the US. This knowledge can be used to maximize urban planning, economic development, and public policy. This knowledge can also prove useful for socioeconomic research when used in conjunction with other datasets such as income distribution for Austin, Texas.

Though the dataset does not have missing data, there are other preprocessing steps that will need to be considered before moving into modeling the data. For example, deciding which features should be kept from the 47 features and if any feature engineering needs to be completed. Scaling the data appropriately since some values are small numbers and others are thousands of square feet or hundreds of thousands of dollars. Potentially transform categorical variables so that a model may apply appropriate weights. This dataset poses unique challenges that will make it a good dataset to work through the data science life cycle towards a model with the goal of accurately predicting housing prices in the Austin, TX area.